

Synthesis for the coordination

PAU Física, UPV/EHU: the ground, the diagnosis, the interpretation, and what follows for 2027 — for the admissions office and for the Bachillerato physics teachers

Josu M. Igartua

20 September 2026

Table of contents

1	The ground	2
1.1	What the data are	2
1.2	What was done with them	3
2	The diagnosis	3
2.1	The size of it, in three frames	3
2.2	Three events, not one	3
2.3	What the 1.65 is made of	5
2.4	What can be subtracted, and what cannot	5
3	The interpretations, and what holds each up	6
3.1	Two things the meeting will ask about	6
4	What cannot be said	7
5	What changes in 2027, and what does not	7
5.1	For the teachers, in one table	7
5.2	Where the reading actually is	8
5.3	The four versions of problem 2, and what they teach	8
5.4	What this means in the classroom	9
6	What we are asking for, and from whom	9
7	What has to be decided, and by when	10
8	Sources	11

! On one page

What happened. The Basque Física mean fell to **3.99** in June 2026, from 5.47 in 2025 and 6.09 in 2024. It is the lowest value in the sixteen-year series and the second consecutive fall.

How much of it is ours. Measured against the nine communities that have published a 2026 result, **1.65 points** of the fall is Euskadi's own and none of the 2025 fall was: in 2025 Euskadi moved with the country to within three hundredths of a point.

How much of that is Física's. Roughly half. The eight Basque subjects and the three that four other communities also published split the change exactly: -1.18 moved all of Eu-

skadi's quantitative subjects together — **Matemáticas II fell 1.67, more than Física** — and —1.10 is Física's own.

What we cannot say. Four fifths of the Basque-specific fall cannot be attributed to anything with the data that has been published, because paper content, reading load, marking severity and newly examined topics leave the same trace in a mean. Separating them needs marks at the level of the sub-task, which EHU holds.

What changes in 2027. Structure, contents and marking do not change. One thing changes: **the statements get shorter**, from about 1 430 words of reading to about 970, with the same verbs, the same rubric and the same demands.

What we are asking for. Three things, in order: the tally of marking deductions for four subjects in 2025 and 2026 (obtainable in days, from inside the university); the sub-task marks of a sample of 2026 scripts; and the tribunal table for Física.

i How to read this document

This is the short form of a longer dossier. Every number here is traceable to a chapter of it, and the chapter is named in brackets. Where a figure is **modelled** rather than observed it is said so on the spot. Nothing here is a causal estimate: these are cohort statistics of different papers marked by different tribunals, and the paper changes every year. The full argument, with its intervals, its failed tests and its two audits, is in [the briefing](#) and in [Chapter 6](#).

1 The ground

1.1 What the data are

Four bodies of evidence, and it matters which is which.

Official statistics. The Ministry's EPAU/SIIU cubes give, for every community and every year from 2015 to 2025, the mean, the pass rate and the band distribution of every subject. They are the only source in which Euskadi and the rest of Spain are measured the same way. The 2026 cube is due in June 2027, so every 2026 figure in this dossier has been assembled from the universities and the regional governments one at a time [Ch. 1](#).

The 2026 results, nine communities. Nine of seventeen have published a 2026 Física mean. Two of them were retrievable only from dashboards that no search engine indexes. The Basque figures come from the EHU deck of 6 July [Ch. 3](#).

The papers. Every exercise and option of the eighteen ordinary papers of 2025 and 2026 in those nine communities — 146 items — was coded on the EHU competency criteria, and a second coder repeated the exercise blind ($\kappa = 0.76$ on the competency flag). The full EHU archive since 2010, 34 papers, was read the same way [Ch. 11](#), [Ch. 12](#).

An instrument nobody here marks. PISA measures the same students at fifteen, on a scale set outside the Basque system, and its 2025 round was published on 8 September 2026 [Ch. 6](#).

What is *not* available is the thing that would settle most of the argument: marks at the level of the sub-task, by tribunal, for the 2026 scripts.

1.2 What was done with them

Everything is recomputed from the raw files by scripts that anyone can re-run; a clean re-run reproduces every table byte for byte and every figure pixel for pixel [Ch. 10](#). The dossier has been audited three times, by readers working only from the files. Each audit withdrew claims the previous version had made, and the audits are published with the rest [Ch. 13](#). That is the reason to trust the numbers below and also the reason to read the section on what they do not establish.

2 The diagnosis

2.1 The size of it, in three frames

Table 1: The same result in three frames [Ch. 6](#).

Frame	What it asks	2026
Year on year	how far did the mean fall from 2025?	−1.48
Against Euskadi's own pre-COVID norm	how far is 2026 from where Euskadi normally sits?	−1.91
Against the nine-community field	how much is ours, once the country's move is out?	−1.65

They span less than half a mark, so nothing here turns on the choice. The field is used for one purpose only, and it is a purpose no absolute number can serve: telling a year in which everyone fell from a year in which only we did.

A one-year fall of 1.48 is rare but not unprecedented — seven of the 170 year-to-year changes in the Ministry panel were as negative. What is exceptional is the **level**: 3.99 is below every regional mean in eleven years of Ministry data except Canarias 2019 [Ch. 2](#).

2.2 Three events, not one

2024 changed the shape and barely touched the level. The mean moved 0.21 — nothing, in a series whose year-to-year standard deviation is 0.85. But the 8–10 band fell nine points where the level predicts three, and the pass rate *rose* two where the level predicts a fall of three. Fewer at the top, more just over the line. As a move of that measure it is the second-most-negative of the 170 transitions in the national panel, and it happened **on the last paper of the old format**, a year before the first LOMLOE paper. Every test in this dossier that works on the mean is blind to it, and all of them are right: a mean cannot see candidates leave the top when others arrive from below.

2025 changed the level, and the change was the country's. Euskadi lost 0.62; the nine-community field lost 0.65. Read against the field there is nothing Basque in 2025 at all.

2026 is ours. The field gained 0.17 and we lost 1.48, so the Basque-specific component is −1.65, and the whole of it is in one year.

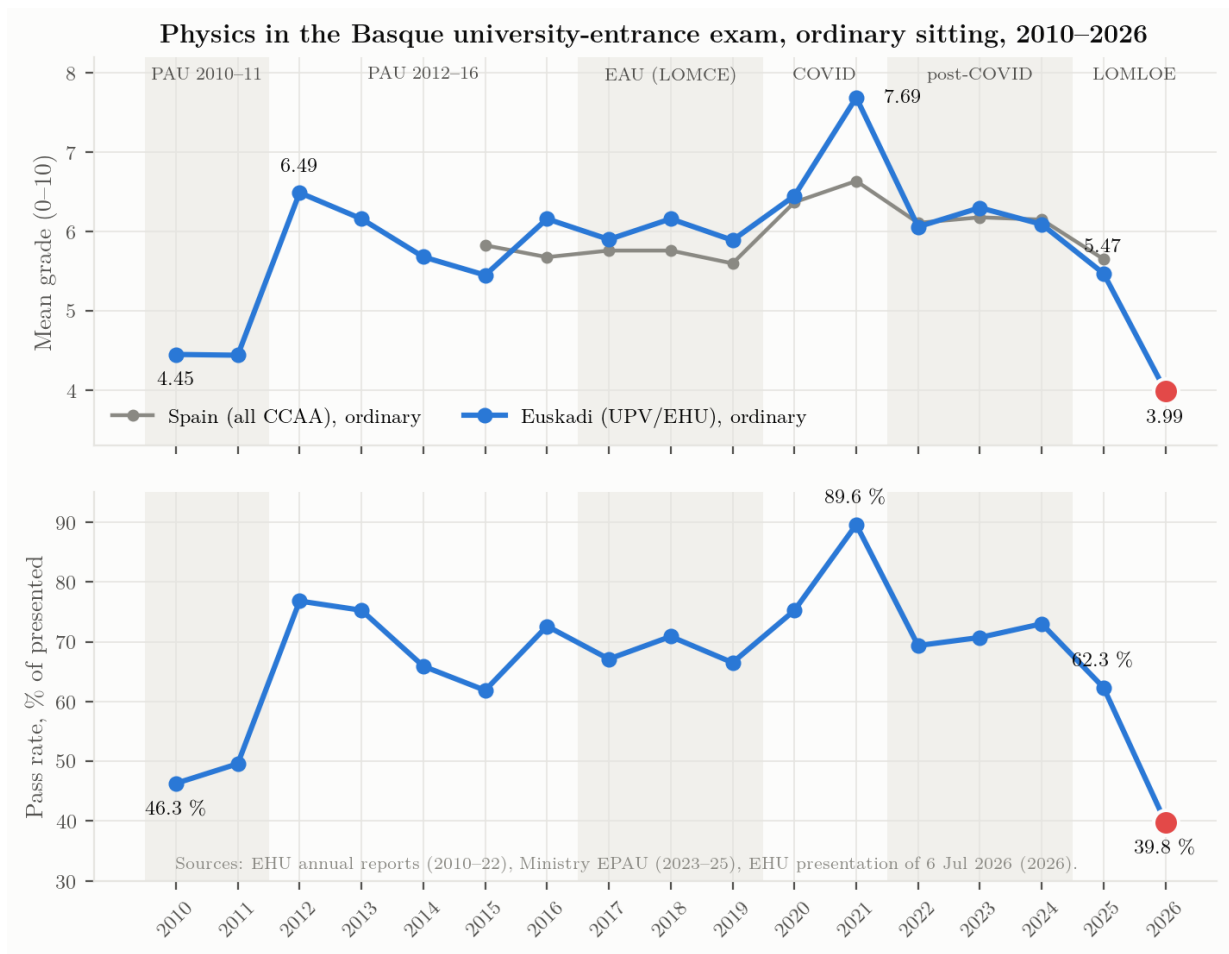


Figure 1: The Basque series 2010–2026 with the exam regimes shaded, and the Spanish total from the Ministry cube in grey.

Figure 26 — Is the 2026 fall Física's, or Euskadi's?

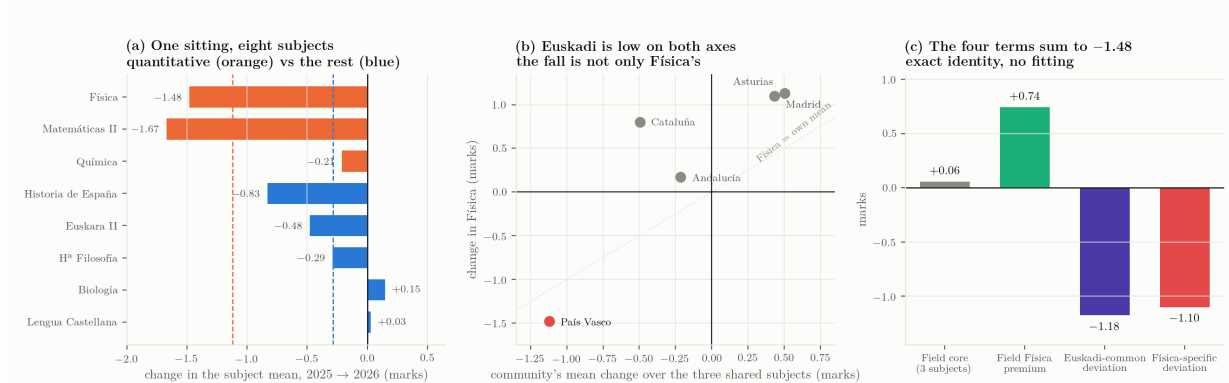


Figure 2: **a** — the eight Basque subjects, with the three the numerical marking deductions can reach in orange. **b** — each community's Física change against its own mean over the three shared subjects. **c** — the four terms of the split, which sum to the observed -1.48 by construction.

2.3 What the 1.65 is made of

Euskadi published eight subjects and four comparator communities published three of the same ones, which gives a balanced panel on which the change splits exactly, with no modelling:

Table 2: The exact split of the -1.48 [Ch. 6](#).

Term	Marks	What it is
Field core	+0.06	what the three shared subjects did in the four comparators
Field Física premium	+0.74	Física above its own community's mean, in all four of them
Euskadi-common	-1.18	what moved all of Euskadi's quantitative subjects together
Física-specific	-1.10	what is left for the Física paper

More than half of what separates us from the field is not Física's. In the same sitting, at the same university, under the same marking specification, Matemáticas II lost 1.67 marks — more than Física — while Química (-0.21), Biología ($+0.15$) and Lengua Castellana ($+0.03$) did not follow.

Two consequences, and both are for the meeting rather than for the dossier. Rewriting the Física paper addresses the -1.10 and cannot touch the -1.18 . And the -1.18 is invisible from inside any one subject's *ponencia*: **Física and Matemáticas II have to be looked at together, and only the office can convene that.**

2.4 What can be subtracted, and what cannot

Table 3: Components of the Basque-specific change [Ch. 6](#).

Component	Marks	Status
Common to the country	+0.17	observed
Choice removed, net of other communities' own cuts	-0.20	quantified (modelled)
Cohort: Basque PISA decline in excess of Spain's	-0.12	quantified (modelled, an upper bound)
Paper content and reading load	—	not identified (one signal seen twice)
Marking severity	—	not identified
Tribunal severity and spread	—	not identified
Newly examined topics	—	not identified (bounded: optional in June)
Who sat the exam	—	not identified (bounded, and it runs the other way)

Two components can be given a defensible size, and both are small: together -0.31 , under a fifth. The remaining -1.33 , **four fifths, is not attributed to anything.** That is the finding, not a gap in the work: the four candidates leave the same trace in a mean, and no further work on published aggregates will separate them.

3 The interpretations, and what holds each up

Table 4: The explanations with verdicts [Ch. 6](#).

Explanation	Verdict	What it rests on
The competency model	Describes 2025 at most; does not explain 2026	Applied in all seventeen communities in both years; six of nine rose in 2026 under it. 2025 was also the end of a plateau present in <i>every</i> subject and cannot be separated from it.
The 2026 paper	Consistent with everything; first-order variable	The whole distribution moved; the damage is in Física and Matemáticas II; it is the only paper of the nine that changed genre, on four axes at once; reading load +45 %; choice halved (worth -0.26 by simulation).
Marking severity	Possible; the <i>timing</i> fits; bounded, not measured	2026 is the second year of the deductions and of the linguistic component, and “cautious in year one, full in year two” predicts exactly the $+0.03 / -1.65$ pattern. But a <i>general</i> severity would move all eight subjects, and Lengua Castellana rose. A severity confined to the seven numerical deductions fits the pattern and is confounded with the format change Ch. 6 .
Tribunal severity	Possible, unmeasured, acknowledged by EHU	The rector reported significant variation between panels in Física; the 6 July deck gave tribunal ranges for three other subjects and not for Física.
A weaker cohort	Real outside the PAU, small inside it	PISA shows a decade-long Basque decline the PAU sees only in Física and only from 2025. But 5.6 % <i>fewer</i> candidates sat Física in 2026, so composition works against the fall, not for it.
Weighting rules make Física dispensable	Plausible slow driver, not a 2026 trigger	Fits a slow erosion, not a fall of a point and a half.
Language or network bias	No evidence either way for Física	Only the Euskara II analysis was run.

3.1 Two things the meeting will ask about

“Was it the students?” Partly, and less than it looks. PISA’s Basque science score fell 21 points between 2022 and 2025, and that is real and measured outside our system. Converted to the PAU scale it is worth about **a fifth of a mark a year** — against a paper-to-paper variation of 0.85 marks. And there is no *rate* to project: the PISA change per cohort-year runs from $+3.7$ to -7.5 , a sequence of steps and not a slope, so a linear extrapolation would be an honest interval around the wrong model [Ch. 6](#).

“Was it the marking?” The timing fits and the subjects bound it. If the tribunals had become generally stricter, Lengua Castellana and Euskara II would show it as clearly as Física; Lengua rose. What the subjects *are* consistent with is a stricter application of the seven numerical deductions — units, vector notation, rounding, SI prefixes, factor of ten, transcription — which can only reach

Física, Química and Matemáticas II, and which fell 0.84 marks further than the rest. One tally of the marking sheets would settle it, and that is why it is the first thing we are asking for.

4 What cannot be said

This section exists because a meeting will over-read a dossier that does not have one.

- **We cannot say the paper caused the fall.** The paper is the first-order variable and the evidence is consistent with it; that is not the same thing.
- **We cannot separate a harder paper from stricter marking.** They leave the same whole-cohort signature, and in 2026 they arrived together — in the same June, in the same community, and in B1 in the *same sub-task*.
- **We cannot say what the -1.18 is.** It is larger than the Física-specific term and nothing in this dossier isolates it.
- **We cannot classify 2026 as a shift or a reshaping,** because EHU has not published the distribution. The two Canarian cohorts that do publish one, at almost exactly our level, are pure shifts.
- **We cannot decide whether 2024 and 2026 are one mechanism or two.** A compression at the top in 2024 and a collapse of the level in 2026 are equally compatible with one slow change in what the marking rewards and with two unrelated events.
- **We cannot project 2027.** The cohort term is a fifth of a mark; the paper is worth five times that.

5 What changes in 2027, and what does not

5.1 For the teachers, in one table

Table 5: What changes and what does not [Ch. 6](#).

	2026	2027
Structure	four problems, 2.50 each	unchanged
Optionality	problems 1 and 2 obligatory; 3 and 4 offer a or b	unchanged
Contents	the four blocks of <i>saberes básicos</i>	unchanged
Rubric	three parts per problem; sub-tasks of 0.25 and 0.50	unchanged
Marking criteria	symbolic first; -0.10 deductions; grave errors void a part; the 0.2 linguistic component	unchanged
Data and constants	one table for the whole paper, with distractors	unchanged
Statement length	about 1 430 words read, 998–1 128 answered	900–1 050 read, 600–700 answered

Shorter is not easier. The verbs, the levels and the rubric are the same; what goes is narrative, repetition between parts, and formatting instructions that the marking criteria already carry.

5.2 Where the reading actually is

Figure 29 — Where the reading is, and what the 2027 target costs

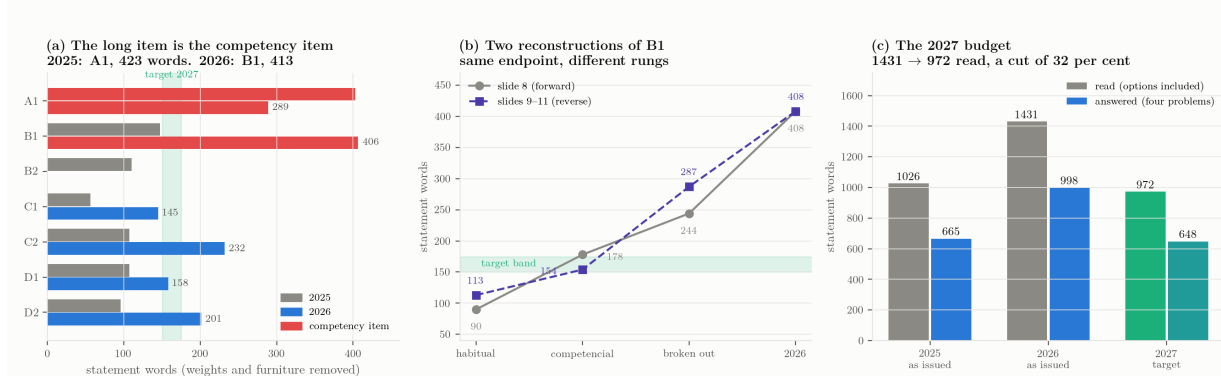


Figure 3: **a** — each item’s statement in 2025 and 2026 against the 2027 target band; the obligatory competency items are in red. **b** — the two reconstructions of the 2026 electromagnetic problem, which share only their endpoint. **c** — what the paper costs to read and to answer, and what the target would cost.

Two facts that only appear when the paper is counted item by item, and that change what “shorter” has to mean.

The length lives in the competency items, and it did in 2025 too. In 2025 six statements ran 56 to 147 words and the seventh — A1, the single obligatory competency exercise — ran 403. In 2026 there are two such exercises, A1 at 289 and B1 at 406, and between them they carry half of everything a candidate reads.

A hundred words of the paper are not questions at all, and they are new. The 2026 paper prints a sub-task or part weight **fifty-one** times, 103 words; the 2025 paper printed none, carrying only the global “2.5 puntos” in its instructions. The granular rubric is an improvement; printing it fifty-one times in the running text is not the only way to deliver it, and a margin column would cost nothing to read.

And the arithmetic the target implies. A competency item, in the two Basque papers that have one, costs 350 to 400 words: A1 was 403 in 2025, and the two of 2026 are 289 and 406, a mean of 348. The four optional items of 2026 average 184. The 2027 budget is 972 words over six statements, and those three numbers do not fit: if the options stay at 184, the two competency items get 118 words each; even bringing the options down to 150 leaves them 186. **The target requires cutting the competency items by 40 to 55 per cent**, from about 350 words to under 200, on precisely the two items that carry the demand the reform introduced. That is why the three rules that change the *architecture* of a competency item — scaffold once, criteria into the rubric, draw the set-up — matter more than the two that trim prose.

5.3 The four versions of problem 2, and what they teach

The 2026 electromagnetic problem is the longest statement in the paper. Its structure is exact and has been checked against the paper line by line: **three habitual calculations asked twice, once per configuration, and three competency elements asked once.** The coordination’s account of *why* — that block B entered the competency paradigm for the first time, so the paper explained, repeated and asked in the habitual way to make the competency demands sit more lightly — is the setter’s testimony, not a verified fact, and it is a presentation of which the author of this dossier is a co-author. What follows does not depend on the account being right.

The coordination has reconstructed the problem in other forms, along two separate routes that share only their endpoint. Going forward from the habitual style: 90 → 178 → 244 → 408 words. Going back from what was issued: 408 → 287 without the narrative, 154 in the compressed form the competency model licenses, 113 written the way the block was examined before the competency requirement, with the set-up as a figure. The two routes are two rewritings, not one rewriting measured twice.

Three things follow for 2027, and the first is the one that changes what is assessed rather than only what is read.

1. **Scaffold once, not twice.** Break the chain out in configuration I and compress it in configuration II. The same support, half the words, and the second configuration then asks the candidate to *carry the method across*, which is the competence rather than a rehearsal of it. None of the reconstructed versions is that form, so it has no measured length; it sits between the 154 of the compressed version and the 287 of the symmetric one.
2. **Keep marking criteria out of the statement.** “Express the result in scientific notation and with two decimals” is not one of the four levels — it is a criterion that applies whether printed or not, and that costs −0.10 when missed. Printed inside the question it adds words, adds no physics, and blurs assessment into penalty.
3. **Write the context as data and draw the set-up.** The 2026 context ran 156 words and says the same in 76; replacing the two prose descriptions of the configurations with two diagrams saves forty more. If the geometry is what you want assessed, *ask for the drawing* instead of printing it — sixteen words, and it becomes a competency element.

5.4 What this means in the classroom

Nothing in the preparation changes because of the length. Four things are worth saying to students anyway, and all four are in the 2026 evidence.

The syllabus is open now. For fifteen years modern physics meant the photoelectric effect — 14 of 15 problems set between 2010 and 2024. Mirrors were set for the first time in June 2025, standing waves with sound levels and carbon-14 dating in June 2026, a mass spectrometer in July 2026. Prepare the blocks as published, not the historical templates [Ch. 12](#).

The deductions are worth counting. Units, a vector arrow missing or misplaced, intermediate rounding beyond 5 %, a wrong SI prefix, a factor of ten, a transcription error: −0.10 each, plus 0.2 for grammar, spelling and presentation on written answers. Five of them is half a mark, and they are cheap to avoid.

Symbolic first, then numeric. It is a marking criterion, not a style preference, and it also protects the candidate: a symbolic answer that is right earns its marks even if the arithmetic goes wrong afterwards.

The constants table brings distractors. Eleven entries in 2026; eight were used and three never. Choosing among them is part of what is assessed.

6 What we are asking for, and from whom

Ordered by what each would resolve, not by how easy it is.

1. **The deduction tally, 2025 and 2026, for Física, Matemáticas II, Lengua Castellana and Euskara II.** How many scripts lost the 0.2 linguistic component, and how many -0.10 deductions per script, by type. *Separates* general marking severity from the quantitative-specific kind. **This is the cheapest decisive request and the only one obtainable in days:** it collects nothing new, the subjects are already marked, and it can be answered from inside the university.
2. **Sub-task marks for a stratified sample of 2026 scripts** (about 300, across tribunals). *Separates all four* unattributed components at once. Run it on B1 first, and run one comparison before any other: **b)'s three sub-tasks against c)'s first, per mark.** Both are the same three-step chain in a different configuration, so if the repetition worked as the scaffolding it was meant to be, c)1 scores above b)1–3; if it was a time tax on the longest statement in the paper, below. The parts must not be compared whole: half of c) is an evaluation sub-task with no analogue in b).
3. **Física by tribunal, 2025 and 2026.** EHU computed this and published the equivalent for three other subjects. *Separates* tribunal severity, and shows whether what is visible in 2026 was already building in 2025.
4. **The full Física histogram, 2024 to 2026,** in one-point bands with the zero and ten counts.
5. **Option take-up in 2026,** and the mean of each option.
6. **July 2026 Física:** presented, passed, mean. **Revision statistics for Física 2026.**
7. **School mark and PAU mark, paired, for the last five cohorts.** The only instrument available for what a cohort gains or loses between fifteen and eighteen.
8. **The other communities' 2026 papers and criteria, and the text that fixes the structural criteria nationally, if it exists.** Until that is settled, the comparison with the nine is a comparison and not a control.

Requests 1, 2 and 3 are the office's to make of EHU. Nothing in the 2027 paper design waits on them; the choice between remedies does.

7 What has to be decided, and by when

Table 6: Who decides what, and when.

	Who	By when
The statement budget and the five rules	the Física coordination	before the coordination instructions go to schools
Física and Matemáticas II read together	the admissions office	as soon as it can be convened
Request 1, the deduction tally	the office, of EHU	now — it is a week's work
Requests 2 and 3	the office, of EHU	before the 2027 paper is written, if they are to inform it
Recording the 2027 paper's parameters before the sitting	the coordination	at the time of setting
A pre-test, double marking, the timetable	the office	budget and calendar decisions

The one recommendation that costs nothing and changes everything about 2027's interpretability: **record the paper's parameters before the sitting** — statement length per problem, compe-

tency share, optional share, penalty rules, topics examined — and read the result against them. Whatever 2027 produces, it will then be a measurement rather than another argument.

8 Sources

Everything above is in the dossier, and every figure in it can be traced to a URL, a page number or a dashboard filter [Ch. 8](#). The dossier's own history, including the three audits and everything they withdrew, is in [Chapter 13](#). The statement analysis draws on the coordination presentation of 18 September 2026 (*USaP Física 2027*, Arretxe, López-Eiguren and Igartua), whose slides 8 to 11 reconstruct the 2026 electromagnetic problem along two routes. Its word counts and this study's agree to within three per cent on the paper total, and from one to six per cent item by item. The author of this dossier is a co-author of that presentation, which is why its design account is treated here as testimony and its counts were recomputed from the archived paper.